Friedrich-Alexander-Universität
Erlangen-Nürnberg

FAU

High End Simulation in Practice

Mini-Project presentation on Smoothed Particle Hydrodynamics

Vinay Kumar Reddy Patil
Aravind Krishnaprasad Saligram

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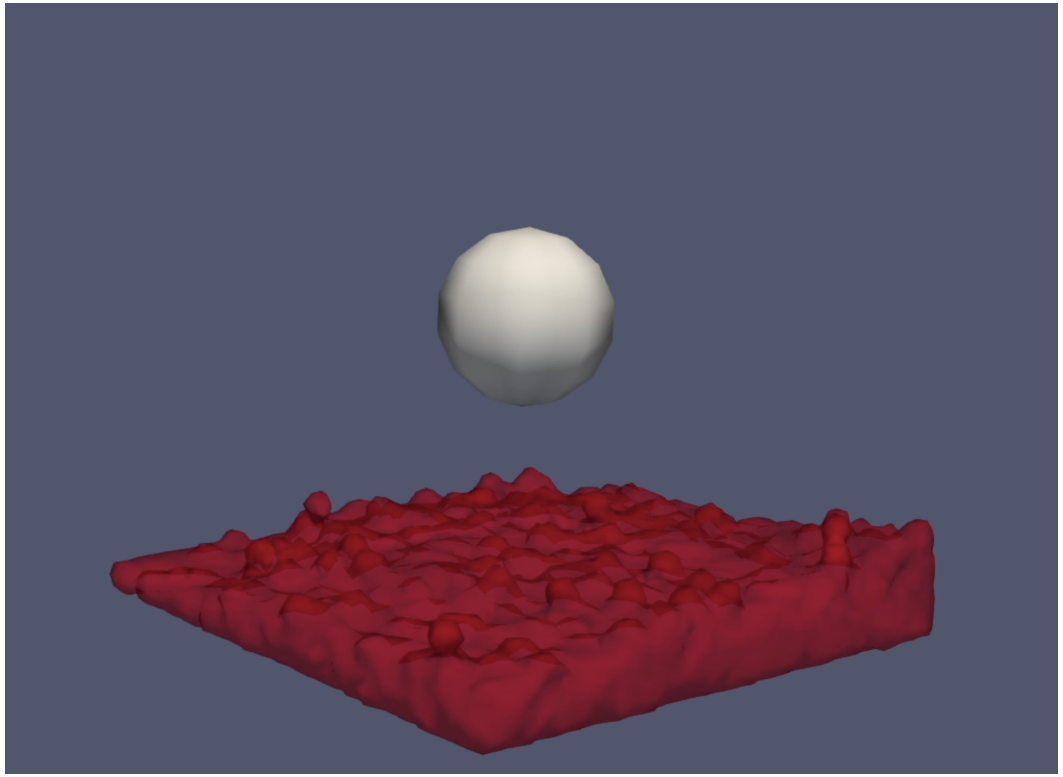
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Introduction

Topic: Creating fluid dynamic simulations using SPH

Abstract: Smoothed Particle Hydrodynamics (SPH) is a Lagrangian method that can be applied to a wide variety of fields. It was created to avoid some of the limitations of finite difference methods. In this mini-project, we aim to implement a GPU compatible SPH solver capable of simulating flow for different scenarios. The boundary condition used in this project is Virtual (Ghost) particles with a hybrid boundary condition.



1. Introduction

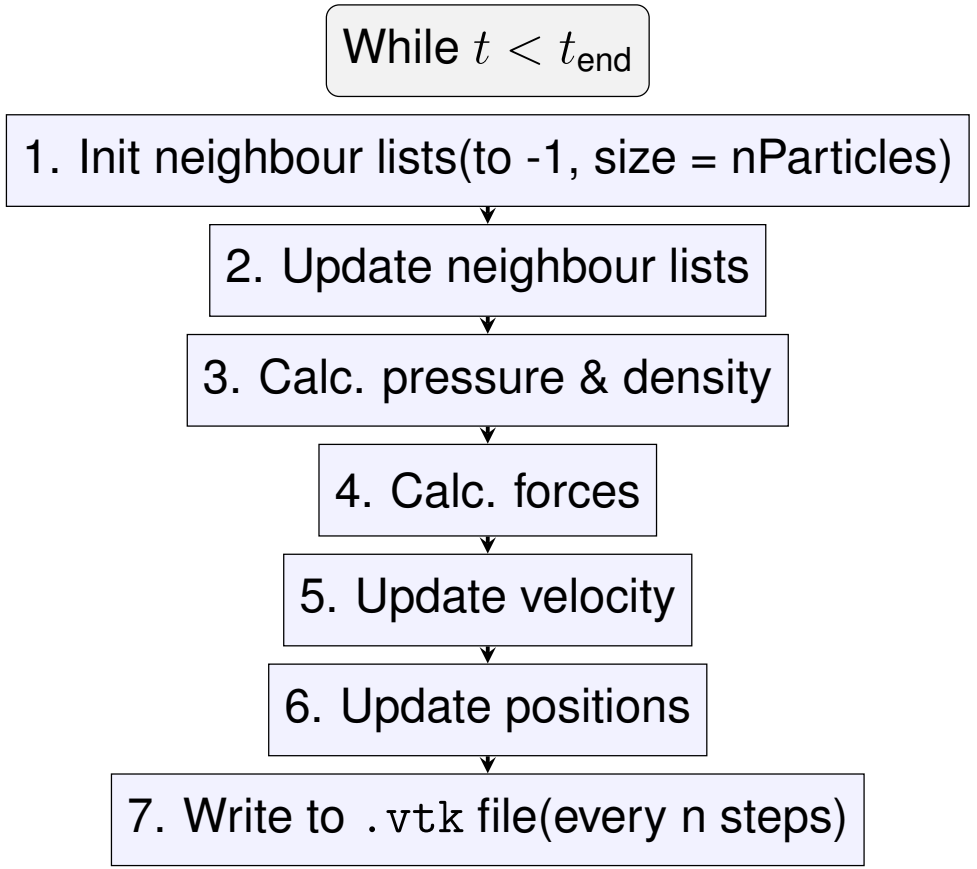
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Methodology

Algorithm



CPU functions:

- Function to create simulation scene
- Function to write `.vtk` files

GPU kernels:

- Kernel to update neighbours
- Kernel to calculate pressure & density of particles
- Kernel to calculate forces
- Kernels to update velocity & position of particles (using Explicit Euler)

Methodology — Governing Equations

Continuity equation:

$$\frac{D\rho}{Dt} = 0 \tag{1}$$

Momentum equation:

$$\frac{D\mathbf{U}}{Dt} = -\frac{1}{\rho}\nabla P + \nu\nabla^2\mathbf{U} + \mathbf{g} \tag{2}$$

Density:

$$\rho(\mathbf{x}) = \sum_j m_j W(\mathbf{x} - \mathbf{x}_j) \tag{3}$$

Pressure (Tait's equation):

$$p = p_0 + k(\rho - \rho_0) \tag{4}$$

Pressure force:

$$\mathbf{F}_i^{\text{press}} = -\sum_j m_j \frac{p_i + p_j}{2\rho_j} \nabla W_{\text{press}}(\mathbf{r}_{ij}) \tag{5}$$

Viscosity force:

$$\mathbf{F}_i^{\text{vis}} = \nu \sum_j m_j \frac{\mathbf{v}_j - \mathbf{v}_i}{\rho_j} \nabla W_{\text{vis}}(\mathbf{r}_{ij}) \tag{6}$$

Gravity force:

$$\mathbf{F}^{\text{gravity}} = \rho\mathbf{g} \tag{7}$$

Methodology — Governing Equations

Kernel gradients:

$$\nabla W_{\text{press}}(\mathbf{r}) = \frac{45}{\pi r_e^6}(r_e - |\mathbf{r}|)^3 \frac{\mathbf{r}}{|\mathbf{r}|} \tag{8}$$

$$\nabla W_{\text{vis}}(\mathbf{r}) = \frac{45}{\pi r_e^6}(r_e - |\mathbf{r}|) \tag{9}$$

$$W(\mathbf{r}) = \frac{315}{64\pi r_e^9}(r_e^2 - |\mathbf{r}|^2)^3 \tag{10}$$

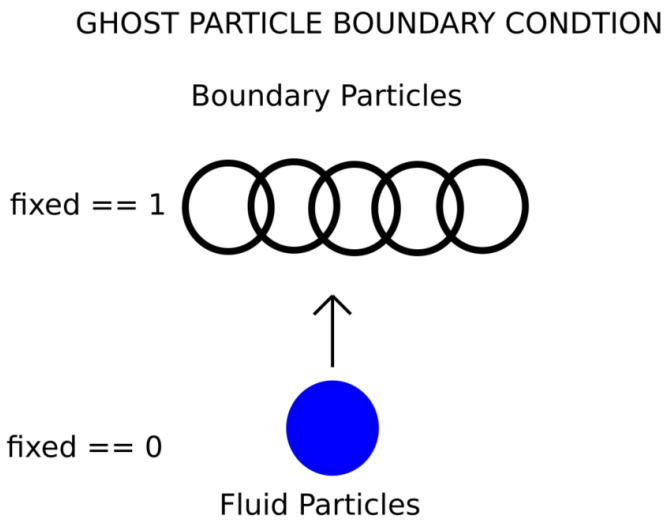
Methodology - Boundary Conditions

- **Different types of boundaries:**
  - Noslip-condition solid boundaries
  - Slip-condition solid boundaries
  - Mixtures of Slip/Noslip
  - Pressure boundaries
  - Flux boundaries
  - Reflective boundaries
- **Different methods to model boundaries:**
  - Boundary particles
  - Ghost particles
  - Virtual forces
  - Analytical methods
- **In current implementation:** Noslip solid boundaries using ghost particles & Hybrid Boundary condition

Boundary Conditions — Ghost Particle Method

Ghost Particle Condition:

- Boundary modeled by stationary particles (`fixed == 1`).
- No position change for boundary particles.



Boundary Conditions — Hybrid Method

Hybrid Boundary Condition:

- Correction only when fluid penetrates boundary.
- Boundary particles stay fixed (`fixed == 1`).
- Fluid velocity is reflected or damped.

Signed Distance Function:

$$F(\mathbf{x}) = \|\mathbf{x} - \mathbf{c}\|^2 - r^2$$

Contact Point:

$$\mathbf{cp}_{\text{sphere}} = \mathbf{c} + r\frac{\mathbf{x} - \mathbf{c}}{\|\mathbf{x} - \mathbf{c}\|}$$

Penetration Depth:

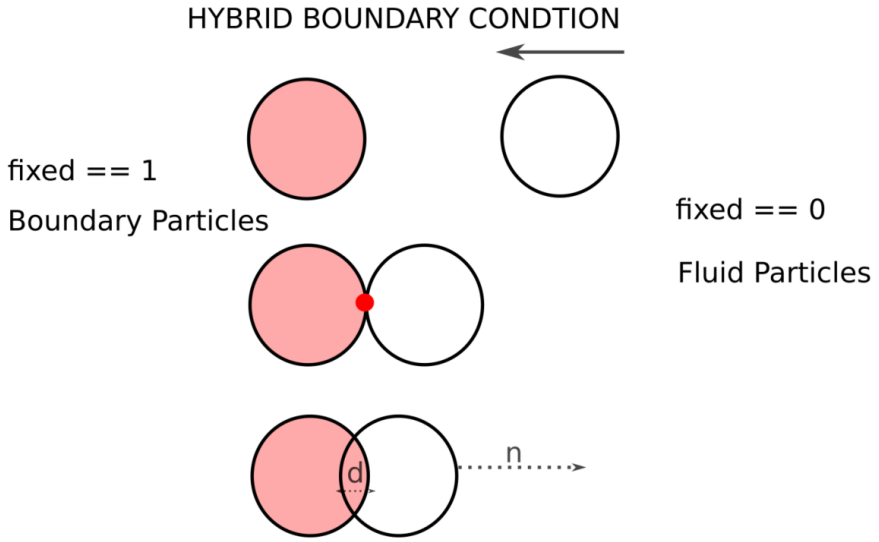
$$d_{\text{sphere}} = \|\mathbf{c} - \mathbf{x}\| - r$$

Surface Normal:

$$\mathbf{n}_{\text{sphere}} = \text{sgn}(F(\mathbf{x}))\frac{\mathbf{c} - \mathbf{x}}{\|\mathbf{c} - \mathbf{x}\|}$$

Abbreviations:

- $\mathbf{x}$: fluid particle position
- $\mathbf{c}$: sphere center, r: sphere radius
- $\mathbf{cp}$: contact point, d: penetration depth, $\mathbf{n}$: surface normal



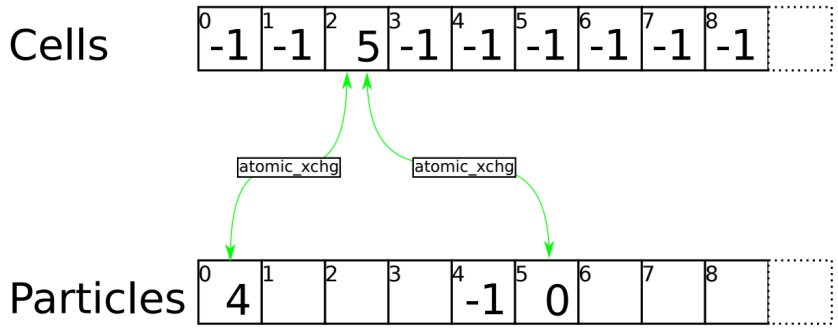
Methodology — Neighbor List Construction

Cell-Linked List for Neighbor Search:

- Each cell stores index of the last inserted particle.
- Each particle stores the index of the previous particle in the same cell.
- Constructed using atomic operations to avoid race conditions.

Atomic Insertion (CUDA):

- Use `atomicExch()` to update the cell head safely.



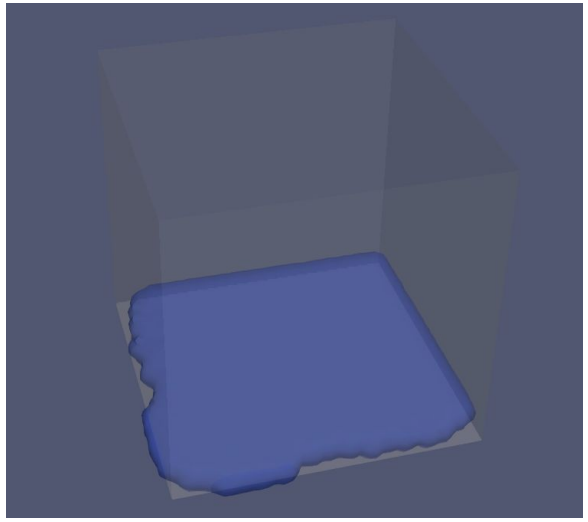
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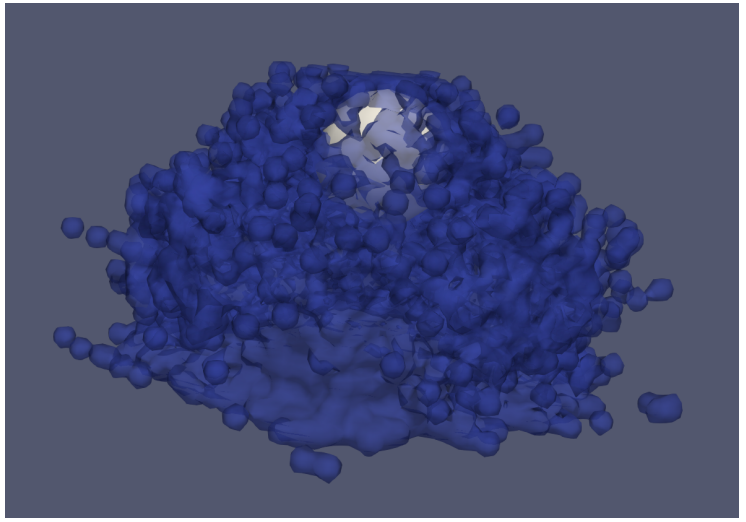
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Scene 1: Free Fall of water



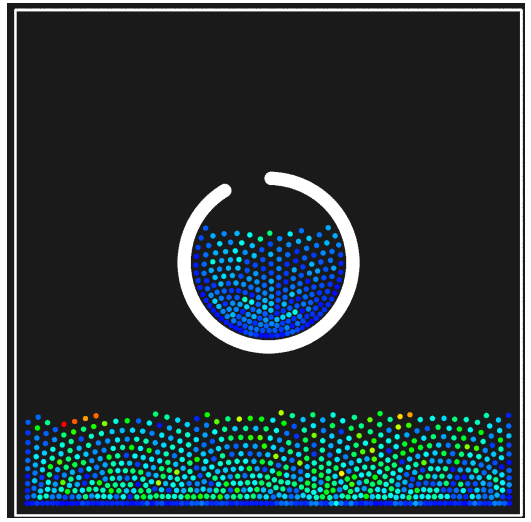
Time step: 0.0001
Total time: 5 sec
fluid particles: 68,921
boundary particles: 540,548
Total: 609,469

Scene 2: Flow over Sphere



Time step: 0.0001
Total time: 5 sec
Fluid particles: 10,000
Boundary particles:: 839,258
Total: 849,258

Scene 3: 2-D Flow in a Cup



Time step: 0.000625
Fluid particles: 1000
Boundary particles: 3927
Total: 4927

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References

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2. Schechter, H. and Bridson, R., 2012. *Ghost SPH for animating water*. ACM Transactions on Graphics (TOG), 31(4), pp.1-8.
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